Computational Fluency Short Course

Version Control and Collaborative Editing

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https://github.com/brownritt/cfsc26

Recap so far: Research *process* supported by tools

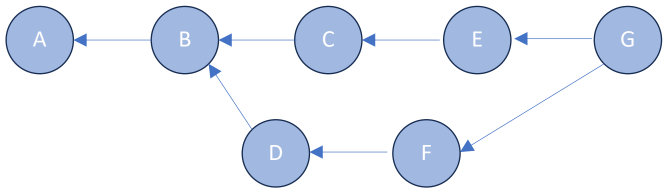
READMEs, specification files	What is the problem definition? What counts as success?
Paths for data and code files	How should I structure my workspace? How do I ensure that I am using the code I mean to?
Virtual environments	What software should I use? How do I make my code reproducible?
Version control	How do I track what I've done? How do I harmonize my work with collaborators?
AI	…?

The tools are just aids for you to take control over and responsibility for your work.

Think about the process first, and then choose the tools to match.

Version Control and Collaborative Editing

Git is the dominant version control system, primarily for text files like code. It saves chains of "snapshots" of the state of files in a "repository" directory, or "**repo**".



There are two main loops in a typical Git workflow:

- **Change-Add-Commit**: Track changes on your local machine
- **Pull-Push**: Synchronize your changes with a "remote" such as GitHub

GitHub is just one of several companies that provide online hosting of Git repos.